



Agentic MobileViT: A stage-wise progressive transfer learning framework with knowledge stabilization for multi-dataset skin lesion classification

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ABSTRACT

Skin cancer has always been a global health concern where early detection and intervention is crucial for active treatment. Skin lesion classification is a foundation of modern dermatology where robust models developed across heterogeneous clinical datasets which still possess challenges in knowledge retention and cross-domain generalization. This paper introduces a novel stage-wise progressive training protocol across heterogeneous datasets rather than strict task-incremental continual learning framework. Our approach uses an innovative MobileViT based hybrid CNN-ViT backbone which symbiotically integrates the local feature extraction with global self-attention to capture both fine-grained and broader contextual cues. The framework improves the state-of-the-art through four key contributions: (i) stage-wise progressive transfer framework that sequentially adapts model parameters across various dermatological datasets; (ii) an agentic memory replay mechanism that adjusts the optimizer dynamically in case of validation stagnation; (iii) the addition of knowledge distillation with transfer learning regularized by replay to achieve representation stabilization via distillation across the training stages; and (iv) computational efficiency analysis representing minimal overhead introduced by the distillation and replay components. The novel feature in the proposed work is its agentic memory replay mechanism which strategically stores a class-balanced subset of samples from prior datasets which ensures that the model retains prior knowledge across training stages while remaining adaptive to new classes. To further enhance the training stability, we introduced a dynamic optimizer switching strategy which wisely adjusts the optimization algorithms and transitions between AdamW and SGD in case of learning stagnation. We rigorously evaluated our model on five heterogeneous skin lesion classification datasets including PH2, HAM10000, ISIC 2019, ISIC 2020, and MILK10k. The model was able to achieve an impressive F1-score of 98.11% and 94.49% on HAM10000 and ISIC 2019 datasets respectively, outperforming existing methods. On clinically focused and challenging ISIC 2020 dataset, the model achieved a high F1-score of 99.24% and AUC of 99.99%. This work is the first to provide a comprehensive performance report on newly published MILK10k dataset, demonstrating that AgenticViT is a powerful adaptive and computationally efficient technique for dermatological image analysis through replay-regularized transfer learning and stag-wise knowledge transfer in real world clinical settings.

1. Introduction

Cancer is one of the principal origins of death globally, with skin cancer being the most frequently diagnosed form of cancer with more than 40% cases annually [1]. Skin cancer typically comprises of Malignant melanoma and non-melanoma skin cancer (NMSC) types, adding to the global burden of cancer [2]. Malignant melanoma originating from melanocytes though less prevalent but usually accounts for most death incidences, hence making it a critical emphasis for healthcare professionals. NMSC usually containing carcinomas on contrary, represents the commonly diagnosed form of skin cancer. Disparities in global incidence and skin cancer burden are evident across countries and regions, largely shaped by population demographics and environmental exposures. Fair-skinned populations with low melanin levels are reported to be most affected by skin cancers [3].

Conventional methods used for detection of various types of skin

cancers include clinical data, visual and histopathological image analysis of skin lesions. During the visual assessment of these lesions, factors such as lesion appearance, location, shape, size, and evolution is taken into consideration. On contrary, histopathological analysis accounts for collecting samples of skin lesion for laboratory examination, typically done through biopsy [4]. To facilitate the examination, dermatoscopes are used. These practices have been effective in skin diseases diagnosis and detection, but can be time-consuming and costly. Computer Aided Diagnosis (CAD) provides Artificial Intelligence (AI) and Deep Learning (DL) solutions based on Convolutional Neural Networks (CNNs) offering an encouraging alternative for early non-invasive and accurate skin cancer diagnosis. Despite the effectiveness of AI and DL methods, AI-based medical assisted diagnosis still faces many challenges such as risk of inaccurate results, volume, scale, depth, data availability, balance of the training data and the computational complexity of AI models [1]. Another main challenge faced by AI based skin lesion classification

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models is the lack of interpretability of the obtained results.

Researchers have used several CNN architectures, like VGG16, ResNet, UNet, Inception and Xception specifically for skin cancer classification. Next generation DL models such as Vision Transformers (ViTs) have shown strong potential for skin cancer classification and diagnosis. These models can handle large amounts of medical imaging data for feature extraction to be used in medical domain but can be computationally expensive as compared to CNNs. Some researchers have explored the combination of these CNN architectures and transformer architectures to achieve better accuracy while keeping a good balance in computation and performance [5,6]. Healthcare applications demand accuracy, which is why the researchers are always working on new advanced AI algorithms for more accurate diagnostic systems.

Many review studies and surveys have analyzed the applications of DL algorithms in skin cancer diagnosis. A comprehensive analysis of existing approaches discussed the potential of advanced Artificial Neural Networks (ANNs), CNNs and hybrid models in effective skin cancer diagnosis [7]. In a recent study, the authors discussed the advances in DL for classification and detection of skin cancer. This study identified that CNNs, ANNs, Radial Basis Function Networks (RBFNs), and K-Nearest Neighbors (KNNs) have mostly been used in literature for skin cancer classification. CNNs surpassed all other methods in image recognition tasks [8]. Another survey reported latest approaches used for skin cancer classification. Pre-trained CNNs such as VGG, ResNet, Inception, EfficientNet, GoogleNet, MobileNet with modifications have been reported, each along with its limitation. Despite a good enough performance by these models, there still remain challenges such as, extensive training which requires powerful devices, variation in skin lesion sizes, less fair-skinned data, less interclass variation in datasets, and data imbalance problem [9].

Current research suggests the potential of ANNs, CNNs and ViTs in accurate detection and classification of skin cancer. However, the challenges and limitations in datasets and model complexity still faced by researchers cannot be ignored. With new subtypes of cancers being diagnosed by dermatologists, with significant morphological similarities, there is a need of new adaptive models that work on the idea of continuous learning and can adapt to new subclasses within the same class. These dynamic models that learn sequentially are vulnerable to catastrophic forgetting, which kills the purpose of introducing them in the first place. Incremental learning enables the models to adapt continuously to new classes without causing interference between recent and prior knowledge. It aims to preserve previous knowledge while adding knowledge from new data. Agentic AI systems are making breakthroughs in AI based healthcare systems by using replay-regularized transfer learning and stage-wise incremental training paradigm. Agentic AI systems are characterized by advanced adaptability, autonomy, scalability and probabilistic decision-making [10]. Agentic AI has the potential to expand the healthcare access and improve the patient outcomes, by improving computational efficiency [11].

While CNNs are computationally competent but struggle to capture long-range global dependencies in image modalities. In contrast, ViTs can capture the long-range dependencies very well but are computationally expensive which makes them impractical for resource-limited environments. The design benefits from the trade-off between CNNs and ViTs which excels the limitations of both and improves the trust medical professionals in automated medical diagnosis through explainability. This study aims to overcome the model complexity, less interclass variation and dataset variability problem by introducing a lightweight vision backbone model using agentic self-adaptive transfer learning paradigm for skin cancer classification. To our knowledge, this is the first agentic model for skin cancer classification achieving notably higher accuracy in comparison to state-of-the-art (SOTA) models. The proposed framework adopts a stage-wise progressive training protocol where multiple skin lesion classification datasets are trained sequentially and exclusively rather than following strict continual learning paradigm. Model parameters learned from one dataset initialize the

training of next dataset. Unlike class-incremental learning where model expands its classifier head to accommodate new classes over time, the proposed model builds a global classifier on all classes across all datasets prior to training eliminating the need for dynamic head expansion. Similarly, unlike task-incremental learning, model does not preserve separate task identities or task-specific inference heads. Our study focuses on progressive cross-dataset transfer stabilization where each dataset is treated as complementary source of knowledge rather than isolated task with a distinct label space. To systematically evaluate the proposed framework, the study is directed by the following research questions:

RQ1: Does stage-wise progressive transfer training improve classification accuracy across heterogeneous dermatological datasets compared to single-dataset training?

RQ2: What is the impact of knowledge distillation on representation stability and classification performance during sequential dataset adaptation?

RQ3: How does replay-based regularization influence model generalization across multiple datasets?

RQ4: What is the computational trade-off (time and parameter complexity) of the proposed Agentic MobileViT framework vs. baseline models?

Contributions of proposed work for skin cancer classification are summarized as follows:

- This work proposes a novel progressive transfer learning framework for skin cancer detection which integrates agentic AI stage-wise learning with the MobileViT architecture. This multimodal approach, progressively transfers optimized model parameters across five diverse dermoscopic datasets (ISIC 2019, ISIC 2020, HAM10000, PH2, and MILK10k). A global classifier is constructed on union of all classes, eliminating the need of architectural modification between stages unlike task and class-incremental learning paradigms. This results in reduced convergence time, minimal representation drift, and accumulation of complementary dermatological knowledge across diverse lesion types and imaging protocols.
- We present an automatic agentic bounded optimization controller that monitors training dynamics and switches between the optimizers in case of learning stagnation. This self-adaptive mechanism improves convergence stability without dataset-specific tuning, with its decisions strictly bounded to dynamic optimizer selection and learning rate adjustment.
- We introduce KD+replay which combines knowledge distillation (KD) from pre-trained EfficientNetB0 teacher with class-importance-weighted memory replay buffer. Replay prioritizes clinically critical classes such as melanoma using weighted sampling, mixed at a 70:30 current-to-replay ratio per epoch. This mechanism ensures representation stabilization across training stages without re-exposing complete data or architectural expansion.
- We present an elaborated computational analysis of the proposed framework. Per-image inference latency and wall-clock training times are reported across all backbone variants under identical hardware conditions, providing a transparent efficiency benchmark against existing methods.
- This research makes a significant contribution to the field by being the first to use the MILK10k dataset, which includes both dermoscopic and close-up images from diverse skin tones, a crucial step toward creating more inclusive and generalizable skin cancer detection models.

Rest of the manuscript is organized for better comprehension and readability. Section 2 focuses on highlighting a thorough discussion on SOTA research associated to deep learning based skin cancer diagnostic systems. A comprehensive description of the datasets and the proposed model architecture is presented in Section 3. Section 4 presents a comprehensive comparison of the proposed AgenticMobileViT model

with SOTA CNN, ViT-based and hybrid models highlighting the experimental results. Discussion and limitations are presented in Section 5. Finally, last section concludes the research highlighting the key findings.

2. Related work

Recent literature highlights the significant contributions made by hybrid architectures combining ViTs with other models like CNN, FCN, U-Net, etc in more accurate skin cancer diagnosis [12]. Yang et al. [13] proposed a novel deep learning model consisting of four-block pipelines responsible for class rebalancing, preprocessing, multi-head self-attention transformer encoder and final convolution. This model accurately classified HAM10000 dataset up to 94.1% by leveraging the weights gained from using transfer learning method on ImageNet dataset. George et al. [14] introduced an innovative five phased model including image preprocessing, segmentation, hybrid optimization based feature selection and extraction, and a BERT based three-tier classifier combined with BiLSTM, Gated Recurrent Unit and Graph Neural Network. Their classification system achieved a high detection accuracy of 97%. Kadirappa et al. [15] proposed a CAD model SASegNet using EfficientNet B1 with multi scale feature fusion. The model applied segmentation first, local and global features are extracted from segmented and original images through EfficientNet B1 model resulting in enhanced AUC-ROC. The proposed model was able to achieve accuracy of 95–96% for binary classification and 91–93% for multi-class classification problem in ISIC datasets.

Skin-CAD, an advanced explainable deep learning based diagnosis system proposed by Attallah for dermoscopic images in skin cancer classification achieved accuracy ranging between 96.2% and 97.2% for two different skin cancer datasets including HAM10000 and Cancer: Malignant vs Benign dataset [16]. Faghihi et al. [17] highlighted the importance of carefully adapting transfer learning framework on fine-tuned pre-trained models to significantly improve the model accuracy. The study uses a merged architecture of VGG16 and VGG19 using transfer learning and achieved the highest accuracy of 98.18% on a subject-specific dermatology dataset. Ayas's work on hybrid CNN and ViT based model benefits by the strengths of both architectures without any preprocessing. The introduced Swin Transformer Model tackles the class imbalance problem by introducing a weighted entropy loss achieving an 82.3% accuracy on ISIC 2019 skin disease dataset [18].

Sasithradevi et al. [19] introduced the idea of using concatenated features and attention modules in EffiCAT to boost the model's accuracy. The model was validated on combined dataset of PAD-UFS-20 and HAM10000 to increase the amount of diverse images for better training samples enhancing the generalization of proposed model with up to 92.08% accurate results. Ahmad et al. [20] proposed a two-stage deep learning approach employing DeepLabv3+ for segmentation and a token-based ViT model for multiple skin cancer datasets achieving high accuracy. Another research proposed a multi-scale long attention network (MSLANet) to address the inter-class similarity challenge. The model contains three long attention networks to improve discriminative representation by introducing a novel depth data augmentation strategy. MSLANet achieved impressive AUC scores on ISIC 2017 and SIIM-ISIC 2020 datasets [21].

Xin et al. [22] introduced an improved ViT-based network named SkinTrans embedding a self-attention module to enhance the important features and repress the noisy features. Authors utilized overlapping and multi-scale sliding windows for image serialization and multi-scale patch embedding to enhance multi-scale features. Contrastive learning was incorporated to improve similar data encoding. The model showed high accuracy on HAM10000 and a private skin cancer dataset

signifying the effectiveness of multimodal transformer based skin cancer classification approaches. Pacal et al. [23] proposed a DL model based on MetaFormer framework, particularly optimized for skin cancer. They introduced a hybrid design integrating focal self-attention mechanism by scaling CAFormer architecture to enhance model's ability to recognize critical regions and improve feature extraction. The model was verified through two benchmark datasets; HAM10000 and ISIC 2019 surpassing existing methods with a light weight design fit for practical clinical use. Soundarya and Poongodi presented a novel approach using hybrid feature extraction and deep learning model. This feature fusion model used Redundant Discrete Wavelet Transform (RDWT), and Gray-Level Co-Occurrence Matrix (GLCM), with pre-trained DenseNet121 model. From various classifiers, XGBoost and ensemble classifier achieved an impressive accuracy of 93.46% and 94.25%, respectively [24]. Aboulmira et al. [25] proposed an innovative hybrid architecture combining wavelet decomposition and EfficientNet models. Integration of wavelet transformations into model's backbone achieved a notable accuracy but increased the computational complexity and processing time. The results of the approach vary depending on image quality and resolution.

Ozdemir and Pacal presented an innovative hybrid deep learning model combining ConvNeXtV2 blocks and separable self-attention mechanisms to enhance feature extraction and improve skin lesion classification accuracy. ConvNeXtV2 captures fine-grained local features while the separable self-attention module ranks the relevant regions performing exceptionally on ISIC 2019 dataset [26]. AbuAlkebash et al. [27] presented a ViT based model that handles the data imbalance challenge by introducing YOLOv8 as preprocessing step. Model's decisions are explained using SHAP values and GRAD-CAM method for better transparency and interpretation. Sargunan et al. [28] presented a DL based technique for skin cancer detection. Their method included a feature extraction mechanism based on Discrete Local Neighborhood Difference Pattern. Segmentation is performed using Recurrent Residual CNN-U-Net (R2U-net) and a hybrid DL model based on Deep Residual Network and Adaptive Neuro-Fuzzy Interface System is used for classification afterwards. Pintelas et al. [29] introduced a feature-augmentation based approach that uses noise-injection functions. Pre-trained ResNet and DenseNet models are used to perform feature extraction. Feature concatenation and dimensionality reduction is done using U-Map. Finally, XGBoost is used for classification achieving an accuracy of 92.9%.

In another research, authors explored the use of CNN and ViT to balance model complexity and performance. The study uses MobileViT for global feature extraction combined with a convolutional layer responsible for acquiring local features [30]. An improved feature representation module is introduced with cosine similarity down sampling module to reduce the information loss [31]. Aishwarya et al. [32] proposed a comprehensive framework experimenting different variations of YOLO. The study revealed that YOLOv5s and YOLOv8s work well on different cancer classes and hence a fusion strategy combining both architectures based on confidence scores is introduced. The results suggest that accuracy improves when the proposed fusion mechanism is employed.

Continual learning (CL) in medical image analysis is also an emerging active research area but is still an underexplored research direction. Bruno et al., conducted a systematic review where 31 researches applying CL to medical imaging ranging from 2018 to 2024 were identified [33]. This lack of research in this area is generally because of lack of standardized benchmarks and inconsistent experimental protocols across the works. Qazi et al., in a survey explored 83 CL methodologies in medical field. Both the reviews converge on three

family taxonomy of CL strategies: regularization based methods, dynamic architectures expanding the network per task and replay based mechanisms that maintain memory buffers for future tasks. The authors further differentiate distillation based regularization as a complementary stabilization technique used for sequential domain shifts [34]. Specifically, in dermatological medical applications, CulpritPrune-Net [35] and Continual-GEN [36] frameworks illustrated that replay based ensemble methods support multi-domain skin lesion classification. Another research explored the naïve fine-tuning based on gradient episodic memory and experience replay for dermatological image classification confirming that replay-based approaches offer a stable knowledge retention mechanism under sequential domain shifts.

Knowledge Distillation (KD) is a technique where a simple smaller student model learns from a complex larger teacher model enabling knowledge transfer using model compression. Chakrabarti et al. [37] implemented a KD framework to generate automated skin cancer reports from dermoscopic images. GPT-4V was used as teacher for generation of pseudo reports and an LLaVA-v1.5-13B student model learned from teacher model to generate clinical diagnostic reports. Saha et al. [38] experimented with four different CNN teacher models and performed knowledge distillation on a simple student model. Out of four, student with DenseNet169 outperformed ResNet150, VGG16 and VGG19. The models were tested on ISIC 2019 and HAM10000 skin cancer classification datasets. Another research implemented a cost sensitive EfficientNetB5 teacher and EfficientNetB2 student model to solve class imbalance problem in ISIC 2020 dataset [39]. Kabir et al. [40] investigated multi-teacher distillation and ensembling methods to improve the performance of student models in KD framework. A stepwise self-knowledge distillation (SW-SKD) framework incorporating feature rectification block and logit rectification block performed efficiently on ISIC 2019, HAM10000, and Dermnet datasets [41]. Elkhayat et al. [42] explored the use of frozen foundation models for continual learning in skin disease classification where a light-weight simple MLP is incrementally trained on a frozen foundation model backbone. Another research used VGG16 and MobileNetV2 with experience replay strategy and achieved a high accuracy of 86.04% with 0.0344 forgetting [43]. Rezk et al. [44] implemented an interpretable skin cancer detection system using the clinical images. Two key interpretability methods are used. First method used domain knowledge to integrate skin lesion classification into model's development and second method uses a visualization method to highlight regions of interest.

The adaptive optimization strategy used in our work is grounded from the Learning to Optimize (L2O) paradigm, where optimizers are learned or adaptively selected on basis of observed training dynamics rather than fixed schedules [45]. In the Automated ML (AutoML) frameworks, learning rate control and adaptive optimizer selection are main components of the automated training pipeline design [46,47]. Our agentic optimizer switch implements a bounded, rule-based instantiation of the same principal. A stagnation counter monitors the validation accuracy per epoch and triggers the optimizer transition (AdamW to SGD) if prolonged stagnation is detected. This provides a reproducible and theoretically grounded adaptive control consistent with lightweight AutoML principles within the constraints of clinical deployment setting while being simpler than fully learned optimizers [48]. However, no existing work applies a unified stage-wise progressive transfer framework across multiple heterogeneous dermatological skin lesion classification datasets under a single global unified classifier with bounded agentic adaptive optimization strategy. Our proposed methodology directly addresses this gap by adopting a priority weighted replay+KD as representation stabilization mechanism, within a progressive transfer learning paradigm that does not require architectural

expansion, task identity, and full dataset re-exposure at any stage.

3. Methodology

This section explains the proposed work introducing a comprehensive stage-wise sequential transfer learning paradigm. This incremental learning pipeline for agentic skin cancer classification is based on teacher-student framework where a pre-trained teacher model provides soft predictions at each incremental stage. These soft labels further guide the student model with MobileViT backbone. Multiple skin lesion classification datasets are trained in progressive manner where the model parameters learned from one dataset initializes the training on subsequent training stages. In this study, all five datasets are combined into a unified training directory prior to model initialization. We constructed a single global unified classifier head with output dimensionality identical to union of all classes across combined datasets, so that both teacher and student model also share identical output dimensions throughout training process. We used Kullback-Leibler (KL) divergence between the teacher and student soft probability distributions to compute knowledge distillation over a consistent class space. No masking, projection or dimensionality reduction is needed since no class-incremental head expansion is performed. The KD objective remains well-defined throughout training. A loss function then combines the ground truth (hard supervision) with teacher predictions (soft supervision) given in Eq. 1.

$$\mathcal{L}_{total} = \alpha \cdot \mathcal{L}_{CE}(y, s(x)) + (1 - \alpha) \cdot T^2 \cdot \mathcal{L}_{KL} \left(\sigma \left(\frac{t(x)}{T} \right), \sigma \left(\frac{s(x)}{T} \right) \right) \quad (1)$$

where y represents the one-hot ground truth, $t(x)$ and $s(x)$ are the teacher and student logits, $\mathcal{L}_{CE}$ represents the categorical cross entropy loss with label smoothing 0.05, $\mathcal{L}_{KL}$ represent the KL divergence with α set to 0.5 balancing the hard vs soft labels and temperature T set to 4.0. This ensures that the student network not only learns new classes but also aligns its feature space with the knowledge distilled by teacher. The proposed model was able to achieve high sensitivity and specificity in skin cancer classification by utilizing light weight vision transformer with agentic AI approach.

3.1. Agentic MobileViT framework

The innovation of this proposed work lies in its capacity to learn from multiple heterogeneous datasets sequentially and exclusively without forgetting previously learned knowledge. Prior to the Stage 1 training, the union of class labels across all five dermatological datasets including PH2, HAM10000, ISIC 2019, ISIC 2020, and MILK10k was computed to define a fixed global output space. This is considered as only an initialization step. The five datasets were then sequentially trained: at stage t , the model trains exclusively on D_t augmented with prioritized replay samples from B_{t-1} . At any prior stage, no data from future stages is accessible. The framework demonstrated in Fig. 1 uses a teacher (EfficientNetB0) and student (MobileViT based hybrid CNN-ViT) distillation strategy which integrates a MobileViT inspired hybrid backbone including an agentic replay memory module and an adaptive optimizer switch. We compute the KD via KL divergence over the same class space eliminating the requirement of output-space realignment, inter-stage architectural modification, or head expansion eventually simplifying the process. Details of complete process are given in Algorithm 1. This section gives a detailed insight to each module of the proposed framework demonstrated in Fig. 2 supplemented with mathematical formulations to thoroughly describe the workflow.

Algorithm 1. Agentic Stage-Wise Progressive Transfer Pipeline for Multi-Dataset Skin Lesion Classification

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1: Ordered sequence of datasets:  $S=[D_1,D_2,D_3,D_4,D_5]$ 
2: Unified class space:  $Y=Y_1 \cup Y_2 \cup Y_3 \cup Y_4 \cup Y_5$ ,  $|Y|$ = total unique classes
3: Initial student model:  $M_0$ , Memory size:  $M = 1200$ , Empty memory buffer:  $B_0 \leftarrow \emptyset$ 
4: for  $t = 1 \rightarrow T$  do
5:   Set current task dataset:  $D_t \leftarrow S[t]$ 
6:   Preprocess and split the dataset: Resize to 224x224, normalize to  $[0,1]$   $D_t^{\text{train}}, D_t^{\text{val}}, D_t^{\text{test}} \leftarrow \text{StratifiedSplit}(D_t, 0.7, 0.2, 0.1)$ 
7:   Load previous student model as teacher:  $T_{t-1} = M_{t-1}$  (frozen; output dim= $|Y|$ ; no masking required)
8:   Load memory buffer from previous stage:  $B \leftarrow B_{t-1}$ 
9:   Compute per-sample priority scores and sample replay batch:
      Priority(x,y) = importancey × (1/classcounty + ε)β (melanoma and clinically critical classes assigned elevated importance scores)
       $R_t \leftarrow \text{ReplaySamples}(B_{t-1})$  using class-prioritized stratified sampling
10:  Construct augmented training set:  $D'_t \leftarrow D_t^{\text{train}}(70\%) \cup R_t(30\%)$ 
11:  Train student: optimize  $M_t$  on  $D'_t$  with KD objective using equation 1 with replay regularization
12:  Evaluate validation accuracy  $val\_acc$  on  $D_t^{\text{val}}$  after each epoch
13:  if  $val\_acc - best\_val\_acc \geq \tau_{\text{opt}}$ , then
14:     $best\_val\_acc \leftarrow val\_acc$ ,  $stagnation\_counter \leftarrow 0$ 
15:  Else
16:     $stagnation\_counter \leftarrow stagnation\_counter + 1$ 
17:    if  $stagnation\_counter \geq 4$ , then
18:      if  $opt = \text{AdamW}$ , then
19:         $opt \leftarrow \text{SGD}(\eta_0 \times 0.1, \text{momentum} = 0.9)$ 
20:         $stagnation\_counter \leftarrow 0$ 
21:      end if
22:    end if
23:  end if
24:  Evaluate  $M_t$  on  $D_t^{\text{val}}$  (report final on  $D_t^{\text{test}}$ )
25:  Update memory buffer with class-prioritized sampling from  $D_t^{\text{train}}$ :
       $B_t \leftarrow \text{UpdateBuffer}(B_{t-1}, D_t^{\text{train}}, M)$ 
      if  $|B_{t-1}| + |new\ samples| > 1200$ , then
        Evict lowest priority samples to maintain  $|B| \leq 1200$ 
      end if
26:  Save  $M_t$  and  $B_t$ 
27: end for
28: Return  $M_t$  and  $B_t$ 

```

3.1.1. Dataset preprocessing and augmentation

Each dataset D_t in the incremental sequence is pre-processed independently before its corresponding training stage t . All input images are resized to 224x224x3 and normalized into the range $[0,1]$. The

dataset is distributed into training, validation, and testing subsets using $D_t = D_t^{\text{train}} \cup D_t^{\text{val}} \cup D_t^{\text{test}}$ using typical ratio of 70:20:10, where we perform stratification over class labels to preserve class distribution across all splits.

A common challenge faced by researchers in skin disease datasets is class imbalance. To handle this imbalance problem, class weights are

used in loss function. The weight per class is calculated as $w_i = N/(C \times n_i)$ where C = number of classes, N = total samples in the dataset, and n_i = samples belonging to class i . Underrepresented classes such as melanoma are automatically assigned higher weights to ensure that their contribution to the loss is proportionally amplified during training process.

Further, data augmentation is applied exclusively to the training sample to avoid overfitting and to improve generalization. Each training image is exposed to random transformations: horizontal and vertical flipping, $\pm 30^\circ$ rotation, $\pm 20\%$ zoom, and contrast/brightness adjustments. The augmented sample set is represented as $X_{aug} = \{T_k(x) | x \in D_t^{train}\}$ where series of various transformations (T_k) are applied to each original image (x) in the sample training subset (D_t^{train}). This pre-processed and augmented dataset is subsequently passed to the MobileViT inspired backbone for stage-wise progressive training.

3.1.2. MobileViT inspired backbone

At each training stage, the student model is initialized with inherited weights from previous training stage, enforcing the stage-wise weight transfer mechanism as described in Algorithm 1. Both teacher (frozen) and student model share identical output dimensionality corresponding to the unified class space. No masking, projection or head expansion is required between stages. Model's backbone is based on MobileViT based hybrid architecture which combines convolutional residual blocks and transformer attention blocks. The convolutional blocks are responsible for local feature extraction and the transformer blocks perform the global contextual reasoning.

3.1.2.1. Convolutional feature extraction. First, a 1×1 convolution is applied to input image of $224 \times 224 \times 3$ resolution trailed by a non-linear activation function σ such as ReLU6 to create z . Pointwise convolution can be represented as $z = \sigma(\text{Conv}_{1 \times 1}(x))$ to expand the number of channels. Subsequently spatial convolution is applied on each channel independently. A 3×3 depthwise convolution is performed with a non-linear activation using $z' = \sigma(\text{DWConv}_{3 \times 3}(z))$.

A final 1×1 convolution is performed and the result is added to the original input x to form the output y . This skip connection helps the model learn incremental feature refinements rather than complete transformations and avoid the vanishing gradient problem supporting stable weight transfer across sequential training stages. Skip connection is omitted if the input and output dimensions do not match.

$$y = \text{Conv}_{1 \times 1}(z') + x$$

3.1.2.2. Transformer attention block. Transformer attention block is responsible for global contextual reasoning. Features are partitioned

into patches of size $p \times p$. The input features are reshaped into tokens. The query, key, and value (Q, K, V) matrices are produced by multiplying the input tokens T with learnable weight matrices (W_Q, W_K, W_V), respectively. Attention score is calculated by taking dot product of query and key matrices and scaling it to the square root of token dimension d_K . Finally, a softmax function is applied to get attention weights. These attention weights are then multiplied by the value matrix to produce the final attention output.

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_K}}\right)V$$

The transformer attention output is then projected back into 2D space and element-wise fused with convolutional outputs to form MobileViT block output. This hybrid representation encodes the long-range contextual dependencies from transformer pathway and local texture patterns from the convolutional pathway, providing discriminative feature space essential for accurate cross-dataset lesion classification. The fused output is then passed to the unified classifier head for final class probability estimation. The trained student model and its extracted features are employed by the Agentic Memory Replay mechanism later on to select and preserve most informative samples for upcoming stages.

3.1.3. Agentic memory replay

Agentic Memory Replay (AMR) mechanism plays an important role in preserving representational knowledge across the sequential training stages. AMR mechanism maintains a structured memory buffer retaining a class-balanced subset of samples from prior training stages as the model progressively adapts to each new dataset. This ensures that the previously acquired feature representations remain available during subsequent adaption without re-exposing full dataset.

A memory buffer M of fixed maximum size stores a representative class-prioritized stratified subset of samples from previously learned samples. Priority is assigned a priority score using importance, x $(1/\text{class}_{\text{count}_y} + \epsilon)^\beta$ where importance _{y} represents the clinical severity of class y and $\text{class}_{\text{count}_y}$ represents the number of samples belonging to that class currently stored in buffer with a smoothing factor ϵ , $\beta = 0.6$ which controls the degree of prioritization. This assigns an elevated importance scores to clinically critical classes such as melanoma to guard against their disproportionate underrepresentation in the buffer sample.

Training distribution containing a combination of current dataset's training samples and replayed samples from M is represented as $D_t' = D_t^{train} \cup R(M_{t-1})$ where $R(M_{t-1})$ are the replay samples in memory buffer of previous stage. Mixing ratio of 70:30 is used for current and replay

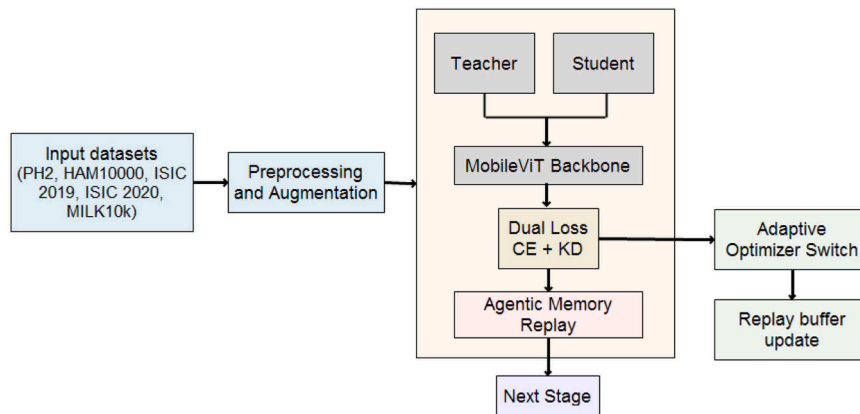


Fig. 1. Flow diagram of proposed Stage-Wise Progressive Transfer Learning Framework with Agentic Optimization and Replay-Regularized Knowledge Distillation for Multi-Dataset Skin Lesion Classification. EfficientNetB0 is used as teacher model and hybrid MobileViT is used as student model with identical output dimensions as the union of all cross-dataset class labels.

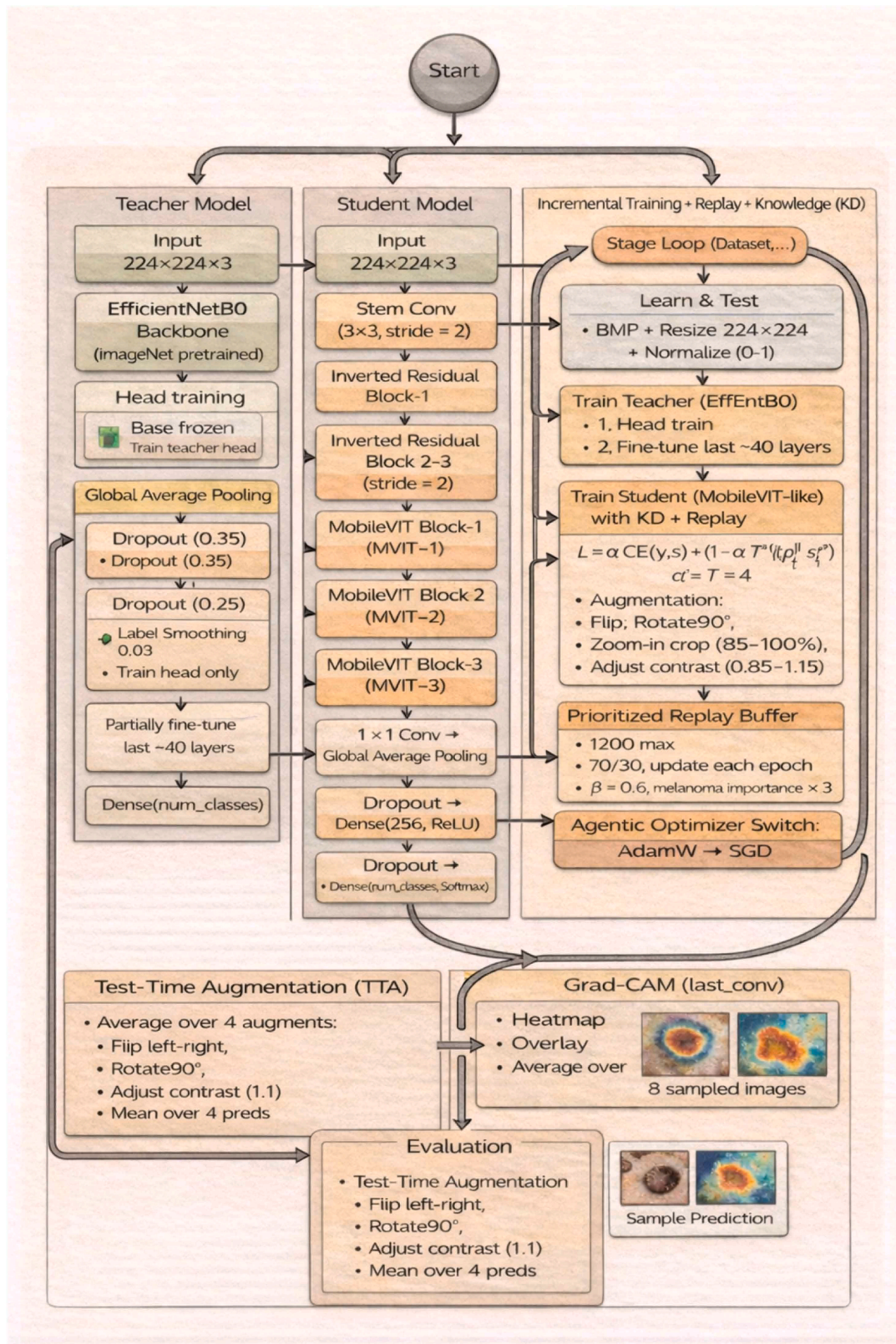


Fig. 2. Proposed Agentic MobileViT Architecture training and evaluation pipeline. The process starts with constructing a foundational MobileViT-based CNN-Transformer hybrid backbone which combines the strengths of convolutional and transformer architectures. During incremental training, new data undertakes augmentation, normalization, and class-weight computation to account for data imbalances. To prevent catastrophic forgetting, current data is merged with a selection of samples from replay memory. The model training is further optimized with callbacks. An Agentic Optimizer Switch dynamically switches the optimizer (e. g., from Adam to SGD or RMSProp) if learning stagnation is detected. Trained model is evaluated using Test Time Augmentation (TTA), and Grad-CAM heatmaps are used to enrich the architecture with explainability and transparency in decisions. For each incremental step, performance metrics are logged.

samples across all experimental stages.

To ensure a balanced training representation across all classes, a stratified strategy in replay selection is used. If the buffer size is $|M|$, for each class c , $\frac{|M|}{C}$ stored samples, the probability of selecting a replay sample during training is:

$$P(\text{replay}) = \frac{|M|}{|M| + |D_t^{\text{train}}|}$$

A lowest-priority eviction policy is applied when the buffer is full to maintain buffer size under 1200. This helps the model to retain samples from rare classes ensuring the buffer contains extremely informative across full dermatological class taxonomy as we introduce new datasets.

The student model's parameters are updated at each training stage using combined loss function considering both current data and memory samples:

$$\theta_t = \theta_{t-1} - \eta \nabla_{\theta} [L(D_t^{\text{train}}; \theta) + \lambda L(R(M_{t-1}); \theta)]$$

Where η =learning rate and λ =replay regularization factor. This combined objective ensures that gradient updates reinforce prior knowledge while concurrently adjusting to the current dataset distribution.

3.2. Bounded agentic optimizer switch

The AMR mechanism functions in combination with the KD and Agentic Optimizer Switch (AOS) to create a united stabilization framework, guaranteeing that the model retains previous representational knowledge while remaining fully adaptive to the current training stage. The AOS is a restricted self-adaptive optimization controller that monitors training dynamics and interferes in case of learning stagnation. This mechanism is applied consistently across all five dataset training stages. Unlike the fully autonomous agents, the decision space of AOS is restricted to optimizer selection and LR adjustment confirming reproducible behavior across all experimental stages.

Validation accuracy val_{acc} is evaluated at the end of each epoch. AOS tracks the improvement in the best recorded validation accuracy $best_{val_{acc}}$ using stagnation counter. When the $stagnation_{counter} \geq 4$, a stagnation event is declared indicating that no meaningful progress was detected for four consecutive epochs. In case of stagnation detection, AOS triggers a transition from AdamW optimizer to SGD with reduced LR promoting fine-grained parameter updates and convergence improvement in later stages where adaptive methods fail to generalize effectively. LR is reset at the start of each new dataset training stage. This dynamic agentic switching ensures robustness against stagnation and overfitting without requiring manual intervention or dataset-specific hyper parameter tuning.

3.3. Explainability

As a formal phase of proposed framework, we incorporate explainability executed independently after each training stage. The Gradient-weighted Class Activation Mapping (Grad-CAM) technique is applied on the final convolutional layer of student model to generate visual explanations on its classification decisions. This ensure model's clinical interpretability and trustworthiness across all five dermatological datasets.

Grad-CAM map for a specific class (c) is calculated by taking a weighted sum of the feature maps (A^k) from last convolutional layer:

$$L_{\text{Grad-CAM}}^c = \text{ReLU}(\sum_k \alpha_k^c A^k)$$

Where the weights (α_k^c) are importance of each feature map for the class c , computed as:

$$\alpha_k^c = \left(\frac{1}{Z} \right) \sum_i \Sigma_j (\partial y^c / \partial A_{ij}^k)$$

This computes the global average of the class score (y^c) gradients with respect to the values of feature map at each pixel position (i,j). This highlights the most significant regions of the image for classification decisions. Regions emphasized using warm tones represent areas of high discriminative importance for predicted lesion class, while the cool colors represent low relevance areas. This spatial localization enables clinicians to verify the alignment of model's predictions with known diagnostic markers such as irregular borders, asymmetry, color variations in melanoma detection to make informed clinical decisions. These heatmaps are generated independently after each dataset training stage allowing monitoring of how model progressively evolves and adapts across heterogeneous datasets providing evidence of stable knowledge transfer across training stages.

The entire workflow of Agentic MobileViT including stage-wise progressive transfer learning, AMR, bounded AOS, KD and Grad-CAM explainability operates as a unified pipeline executed incrementally across all five datasets where each component compliments others. The backbone extracts discriminative hybrid representations, AMR preserves cross-stage knowledge, AOS guarantees convergence stability, KD objective anchors representation learning and Grad-CAM provides transparency in clinical interpretability, which all together provide a computationally efficient, progressively adaptive, interpretable framework for real-world dermatological image analysis.

4. Results and discussions

This section comprises the details of selected dermatological classification datasets, including data preprocessing and augmentation, evaluation metrics, experimental setup, results and comparison to different backbones.

4.1. Skin disease datasets

In this study, we have used popular publically available PH2, HAM10000, ISIC 2019, ISIC 2020 and MILK10k skin cancer datasets. A summary of these datasets is stated in Table 1.

PH2 dataset contains 200 dermoscopic skin images with three classes; atypical nevi (80) and common nevi (80), and melanoma (40) [49]. HAM10000 comprises of 10015 high resolution dermoscopic images with seven classes; melanoma (mel 1112), melanocytic nevi (nv 6705), benign keratosis-like lesions (bkl 1098), basal cell carcinoma (bcc 514), actinic keratosis (akiec 327), vascular lesions (vasc 142), and dermatofibroma (df 115) [50]. International Skin Imaging Collaboration (ISIC) is a global partnership that maintains a large, publically accessible repository of dermoscopic skin cancer images. ISIC is a primary resource for researchers working on ML models for skin cancer analysis. ISIC 2019 is a repository of 33569 dermoscopic skin lesion images separated into 9 classes; melanocytic nevus, melanoma, dermatofibroma, vascular lesion, basal cell carcinoma, squamous cell carcinoma, actinic keratosis, benign keratosis, and none of the others. 25,331 images are available for training across 8 classes. ISIC 2019 is known for its class imbalance challenge with some lesions more frequent than the others. In contrast, ISIC 2020 addressed a critical limitation of previous skin lesion image datasets by focusing on patient-centric approach. It includes multiple images from each patient. Main focus of researchers using ISIC 2020 was to identify histologically confirmed cases of melanoma. ISIC 2020 contains 33,126 training and 10,982 testing dermoscopic skin lesion images categorized into 6 classes [51]. MILK10k is a latest and hence rarely explored skin disease dataset expanding upon existing skin datasets with further enhancements. MILK10k consists of 10480 images with pairs of dermoscopic images and clinical close-ups of 5240 lesions distributed over 11 classes [52]. In addition to 8 similar classes to ISIC 2019 and HAM10000, three additional classes are introduced; infectious and inflammatory conditions, other malignant and benign propagations including collision tumors. MILK10k dataset is the first dataset containing image from diverse skin tones ranging from very dark to very

Table 1

Public skin disease datasets stats.

Dataset	No. of classes	Total No. of images	Image type
PH2	3	200	.bmp
HAM10000	7	10015	.jpeg
ISIC 2019	9	33569	.jpeg
ISIC 2020	6	44,108	.jpeg and DICOM
MILK10k	11	10480	.jpg

light which makes it the first benchmark dataset containing pigmented skin images unlike existing datasets.

4.2. Evaluation metrics

Model's performance is evaluated using standard classification metrics. Typically accuracy (ACC), precision (P), recall (R), F1-score (F1), and ROC-AUC (area under the receiver operating characteristic curve) are used for evaluation of classification tasks.

$$ACC = \frac{TP + TN}{TP + TN + FP + FN}$$

$$P = \frac{TP}{TP + FP}$$

$$R = \frac{TP}{TP + FN}$$

$$F1 = \frac{2PR}{P + R}$$

Additionally, Test Time Automation (TTA) is used to improve robustness during evaluation, by averaging predictions across multiple augmented versions of the same image.

$$\mathcal{Y} = \left(\frac{1}{K} \right) \sum_{k=1}^K f_{\theta}(T_k(x))$$

Where $\mathcal{Y}$ is the final prediction vector, $T_k(x)$ represent model f_{θ} 's prediction on k^{th} augmented version of input image x .

4.3. Experimental setup

Proposed model was trained on Intel Core i9 CPU and NVIDIA GeForce RTX 4060 GPU with Windows 10 using TensorFlow/Keras framework. For fair and reproducible assessment, all baseline models were also evaluated on identical hardware under identical software configurations. The framework incorporates replay memory and KD inspired by continual learning literature, the experimental protocol follows a stage-wise progressive transfer learning paradigm rather than following a strict sequential task-incremental learning. Selected datasets are trained progressively where for each next stage, model weights are initialized with optimal weights from previous stage. We conducted the evaluation independently on the held-out test split after each stage. Since the model avoid re-visiting earlier datasets for evaluation after subsequent training stages, we do not compute formal continual learning retention metrics such as forgetting scores and backward transfer in our study. The hyperparameters used uniformly across all five dataset training stages are recorded in Table 2.

The dataset splitting protocol of 70:20:10 was used for splitting the datasets into training, validation and testing subsets, respectively. To preserve the original class distribution across all three splits, stratification is performed based on class labels. We conducted the splitting at image level in our current study rather than enforcing patient-disjoint separation. This may introduce a potential patient-level correlation between splits. We evaluate all models under the same protocol to ensure valid ablation comparisons. We conducted the evaluation independently on the held-out test split after each stage. Since the model avoid re-

visiting earlier datasets for evaluation after subsequent training stages, we do not compute formal continual learning retention metrics such as forgetting scores and backward transfer in our study. The hyperparameters used uniformly across all five dataset training stages are listed in Table 2. The dataset splitting protocol of 70:20:10 was used for splitting the datasets into training, validation and testing subsets, respectively. To preserve the original class distribution across all three splits, stratification is performed based on class labels. We address the class imbalance problem using cross-entropy loss function at each training stage ensuring that the underrepresented classes contribute equally to parameter updates without need of synthetic data generation and oversampling. These adaptive rules unified in our framework are applied consistently across all five datasets.

4.4. Experimental results and visualization analysis

Proposed model performance was tested on five skin disease datasets in sequence: PH2, HAM10000, ISIC2019, ISIC2020, and MILK10k. Performance was assessed on training, validation and test splits for each dataset (Table 3). Model demonstrated exceptionally high performance on training data almost across all datasets with accuracy and F1 score consistently reaching 0.99 except for 0.96 for MILK10k dataset. This shows that the model effectively learned and fitted the training data for each respective dataset. Generalization capabilities of the proposed framework were assessed on validation and test datasets. For PH2, HAM10000, and ISIC2020 datasets, the model maintained a strong performance on test datasets with ACC, F1, and AUC values consistently above 0.98 demonstrating the model's robust performance on these datasets. Model's test performance was slightly lower than training performance on ISIC 2019 dataset suggesting a minor performance drop while generalizing to unseen data. Another drop in performance was seen during testing of MILK10k dataset with training scores above 0.96 and ACC and F1 scores dropped to 0.87 while AUC remains at 0.99. This performance gap, particularly visible in MILK10k dataset highlights the importance of model evaluation on diverse and challenging datasets to identify potential limitations in generalization. Performance gap in train and test ACC, P, R, and F1 metrics indicate that while the model has learned to rank instances correctly, the optimal threshold for classification might differ between training and test sets. High AUC scores achieved for all datasets is a reliable indicator of our model's discriminatory power.

Fig. 3(a-e) demonstrates the confusion matrices highlighting detailed outline of class-wise performance highlighting the effectiveness and robustness of proposed model on all five datasets.

The proposed framework was able to achieve a weighted average precision, recall, and F1-score of approximately 99% across 384 images of PH2 dataset. Model showed excellent diagnostic performance for NV class with a recall of 1.0000, precision of 0.9789 and F1-score of 0.9893 indicating that the model classified all 139 images of NV class correctly. Model also demonstrated strong performance for MEL (melanoma), a critical type of skin cancer with precision of 1.0000, recall of 0.9911 and

Table 2

Hyperparameters across all five dataset training stages.

Initial Learning Rate (LR)	1×10^{-4}
SGD Learning Rate (upon AOS)	$\eta_0 \times 0.1 = 1 \times 10^{-5}$
SGD Momentum (upon AOS)	0.9
Batch size	24
Maximum epochs per stage	100
Replay buffer size	1200 samples
Replay batch size per epoch	300 samples
Replay priority factor (β)	0.6
Current to replay mixing ratio	70:30
Distillation temperature (τ)	4
Distillation weight (α)	0.5
Optimizer stagnation patience	4 epochs
Improvement threshold	1×10^{-4}

Table 3

Experimental results on five skin cancer datasets.

Split	ACC	P	R	F1	AUC
PH2					
Train	0.9971	0.9971	0.9971	0.9971	0.9999
Val	0.9836	0.9840	0.9836	0.9836	0.9986
Test	0.9899	0.9898	0.9896	0.9896	0.9998
HAM10000					
Train	0.9984	0.9984	0.9984	0.9984	0.9999
Val	0.9858	0.9859	0.9859	0.9858	0.9993
Test	0.9811	0.9812	0.9811	0.9811	0.9988
ISIC2019					
Train	0.9910	0.9910	0.9910	0.9910	0.9999
Val	0.9464	0.9469	0.9464	0.9464	0.9958
Test	0.9448	0.9454	0.9448	0.9449	0.9959
ISIC2020					
Train	0.9937	0.9938	0.9937	0.9937	0.9999
Val	0.9920	0.9921	0.9920	0.9921	0.9999
Test	0.9924	0.9926	0.9924	0.9924	0.9999
MILK10k					
Train	0.9616	0.9616	0.9616	0.9614	0.9987
Val	0.8754	0.8774	0.8754	0.8750	0.9868
Test	0.8706	0.8736	0.8706	0.8703	0.9860

F1-score of 0.9955 emphasizing its robust generalization and high accuracy of critical diagnosis.

For HAM10000 dataset, the model demonstrated weighted average accuracy, precision, recall and F1-score of 98% indicating that despite class imbalance, the model performed exceptionally well consistently. Model achieve a perfect F1-score across df and vasc classes correctly classifying all 1052 and 983 images. Model reports strong results for akiec and bcc classes with 0.99 F1-score. For bkl and mel, the f1-score

was reported 0.97 whereas it was reported 0.96 for nv class showing a balanced tradeoff between precision and recall for this class.

The model's performance on ISIC 2020 notably improved as compared to ISIC 2019 dataset with accuracies reported as 99.2% and 94.5%, respectively. Model's superior performance on ISIC 2020 in comparison to 2019 is directly attributed to the characteristics of images within both datasets. ISIC 2019 is a diverse and large collection of skin lesion images from different origins. This heterogeneity in large number of classes consisting of images with different quality, lightening, magnification and patient demographics introduces significant challenge for classification model which is reflected in the model's lower F1-scores for some classes. On the other hand, ISIC 2020 was specifically designed to be more clinically relevant. Model achieved a perfect F1-score of 1.00 for bkl, df, nv, and vasc on ISIC 2020 dataset. Model's consistent high performance on this dataset suggests that its advanced architecture and memory-based training framework were able to capitalize on the improved consistency, quality, and contextual richness of the ISIC 2020 dataset.

Classification report for MILK10k dataset shows that the model attains overall high accuracy while achieving varied performance across individual classes. Model achieved a weighted accuracy of 87.06%, which is still strong given the heterogeneity and complexity of the dataset. The consistent values of macro and weighted averages indicate that the model is not biased towards any single class.

Table 4 highlights that our proposed framework constantly outperforms all the latest models published in 2025 across multiple skin lesion datasets. The results from HAM10000 and ISIC 2019 demonstrate that the proposed unique combination of incremental learning with AMR ensures high accuracy and F1-score as compared to other mode

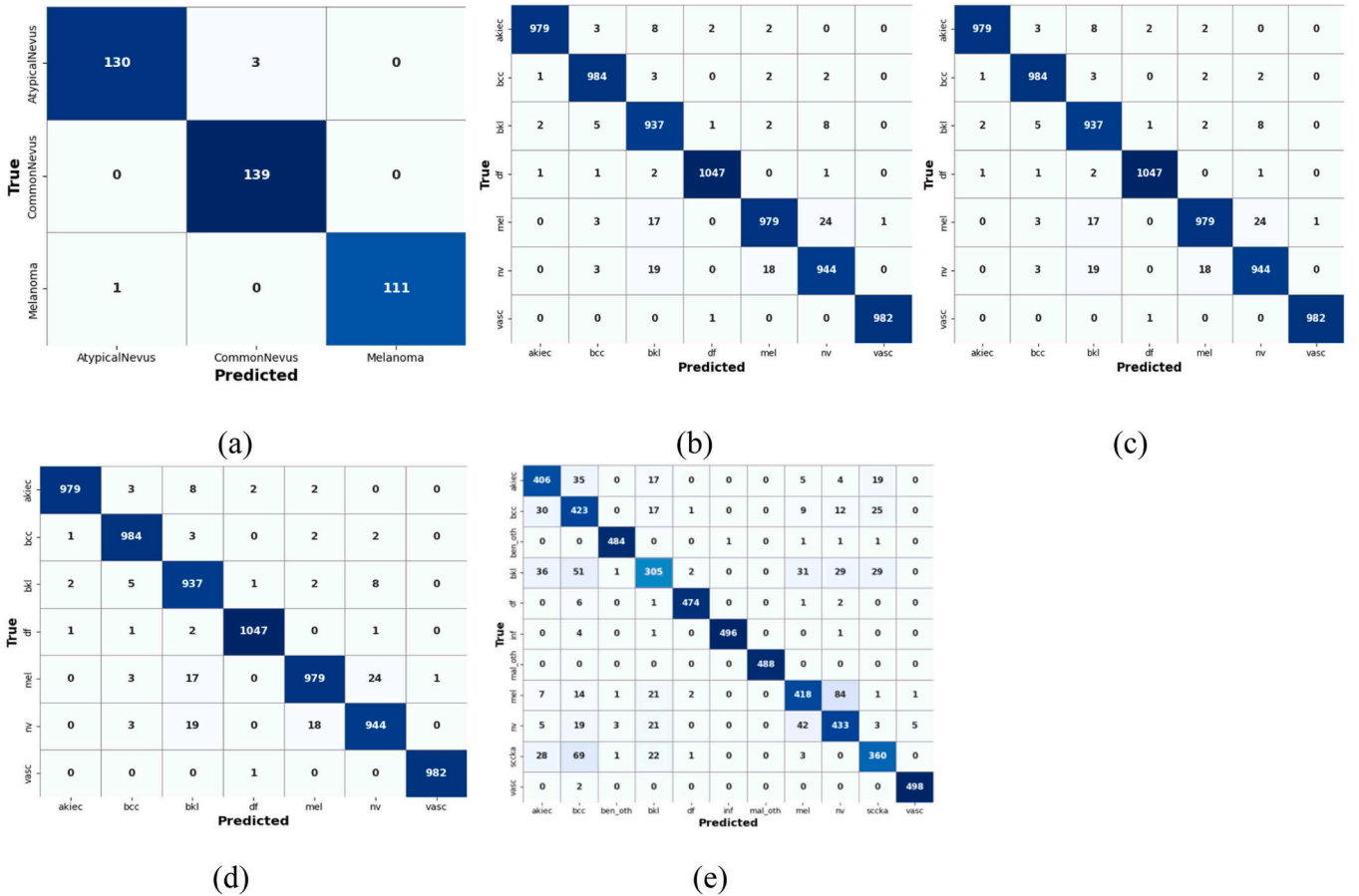


Fig. 3. Normalized Confusion matrix demonstrating class-specific performance of proposed Agentic MobileViT framework on test sets of five benchmark skin cancer datasets after incremental learning (a) PH2 (b) HAM10000 (c) ISIC2019 (d) ISIC2020 (e) MILK10k.

including simple KD approaches or similar hybrid architectures. F1-score of 98.11% shows the model's ability to effectively handle complex image variations and maintain high performance across all classes. Similarly, F1-score of 94.49% outperforming all SOTA models exhibits model's strong generalization capabilities on a highly diverse and challenging dataset. The proposed model achieved impressive performance for ISIC 2020 dataset, with perfect AUC of 99.99% and F1-score of 99.24%. These results are an evidence to the model's advanced design which allows it to excel in clinically representative datasets proving the effectiveness of integrated KD with AMR and a dynamic training strategy. Furthermore, a direct comparison of results of MILK10k datasets is not possible at the moment as the dataset is recently introduced with almost no research work available on it yet. Our work is first to provide a comprehensive performance report on MILK10k dataset establishing a valuable benchmark for future research.

Figs. 4 and 5(a-e) represent the ROC (Receiver Operating Curves) and Training curves showing the loss per epoch for all five datasets, respectively.

To further validate the interpretability and reliability of the proposed model, Grad-CAM is applied to visualize the most important regions in the model classification decisions. Grad-CAM results provides explainability to the framework which is essential in medical imaging to further provide confirmation that the model is focused on clinically relevant features. By overlaying the Grad-CAM heatmaps over original images, we can analyze whether the activated regions correspond to lesion structures typically considered important by dermatologists, such as the lesion core, irregular borders, and, pigment variations.

The results in Fig. 6 show that the model consistently attends to diagnostically and clinically meaningful regions. In the first case, heatmap is concentrated along lesion boundary showing that the network emphasizes shape irregularities that often guide clinical assessment. In the second case, the highlighted red-yellow regions are distributed within lesion core and along its edges. This reflects attention to both internal texture and border definition. Overall, the Grad-CAM visualizations confirm that the model prioritizes lesion-specific features and is paying attention to lesion's core and boundaries, and is not distracted by

irrelevant background patterns. Such behavior strengthens confidence in the model's predictions and underscores its potential utility as an explainable reliable diagnostic aid in dermatology making trustworthy predictions.

4.5. Ablation studies

This section exhibits the results of ablation studies conducted to examine the impact of different backbones and architectural choices within the proposed Agentic framework. All ablation experiments were performed under identical hardware, hyperparameter configurations and data splitting protocols to ensure reproducible and fair comparison. This section is structured in three sections: (i) comparison of backbone architectures, (ii) analysis of KD contribution on PH2 dataset, and (iii) analysis of computational efficiency across all model variants.

Table 5a presents the results of backbone comparison across all five dermatological classification datasets under stage-wise progressive transfer learning protocol. The results demonstrate that the Agentic MobileViT architecture consistently outperforms the Swin Transformer and EfficientNetViT backbones across all datasets achieving outstanding test accuracies ranging from 94% to 99%. On the contrary, Swin and EfficientNetViT achieved significantly lower test accuracies, typically ranging between 65% and 75%. This gap between the reported accuracies indicate that the traditional transformer and CNN architectures have limited generalization and adaptability whereas our proposed framework has the ability to extract highly discriminative features when applied to progressive multi-dataset training conditions.

On the ISIC 2019 and 2020 datasets, proposed model using MobileViT backbone attained above 94% accuracy whereas Swin and EfficientNetViT were restricted only to 67–75%. This margin suggests that the proposed MobileViT's lightweight hybrid CNN-ViT architecture is more effective in modeling complex lesion distributions under class-incremental scenarios. Even on large-scale and more challenging datasets like MILK10k, Agentic MobileViT was able to sustain competitive test accuracy of 87% which remained up to 70% using competing backbones, highlighting the superior robustness of the proposed model. Inference latency measured as ms/image averaged over warm-up and time forward passes on identical hardware.

These outcomes address RQ1 and further suggest that the integration of local convolutions with global self-attention mechanism in MobileViT leads to more expressive feature representations. Such representations enable the model to both generalize to unseen classes and retain prior knowledge during incremental updates. In contrast, the competitive rigid and complex architectures of Swin and EfficientNetViT appear less amenable to fine-tuning under replay-driven training dynamics, which likely contributes to their lower performance.

Table 5b presents the quantitative computational efficiency analysis across all model variants on PH2 dataset. The formal time complexity of proposed work is organized across three components. A dominant cost of $O(N(HwK^2d + Ld^2))$ is contributed by the backbone where the local patch-based tokenization in MobileViT bounds transformer sequence length, keeping Ld^2 (the attention cost) manageable. KD adds $O(NC)$ at each training step where replay strategy contributes to a fixed overhead of $O(R)$ independent of the dataset size. Therefore, the total training complexity becomes

$$O(N(HwK^2d + Ld^2)) + O(NC) + O(R) \approx O(N(HwK^2d + Ld^2))$$

ensuring that the KD+replay does not change the asymptotic complexity beyond the backbone cost.

The space complexity of proposed framework comprises $O(P)$ for the model parameters and $O(R.HW)$ for replay buffer. These components remain practically bounded with $P = 1.68M$ and $R = 1200 < N$ across all five datasets.

Student model which is a hybrid MobileViT achieves a 93.0% parameter reduction compared to ResNet50 and 61.7% parameter reduction compared to EfficientNetB0 architectures, directly addressing

Table 4
Comparison of proposed framework with 2025 SOTA methods.

Ref#	Methodology used	Accuracy (%)	F1-score (%)
HAM10000			
[23]	CNN-ViT	95.01	93.34
[25]	Integrated wavelet transformations with an EfficientNet backbone	94.70	95.00
[27]	ViT model integrated with YOLOv8	93.00	88.00
[31]	Hybrid CNN-ViT	97.83	97.81
[42]	PanDerm with MLP-based CIL and Prototype-based Nearest Mean Classifier	92.25	-
[38]	KD approach with DenseNet 169 teacher backbone	93.24	-
[40]	KD multi-teacher with Average concatenated ensemble TinyStudent	88.00	-
[53]	Modified Long Short-Term Memory (MLSTM)	96.20	95.10
	KD framework with agentic memory replay (Ours)	98.11	98.11
ISIC 2019			
[23]	CNN-ViT	92.54	88.86
[26]	Transfer learning+ConvNeXtV2 with self-attention	93.48	91.82
[31]	Hybrid CNN-ViT	83.29	83.45
[38]	KD approach with DenseNet 169 teacher backbone	84.13	-
	KD framework with agentic memory replay (Ours)	94.48	94.49
ISIC 2020			
[32]	Integrated predictions from YOLOv5s and YOLOv8s	-	97.00
[39]	KD framework with EfficientNet-B5 teacher and EfficientNet-B2 student model	92.95 (AUC)	-
	KD framework with agentic memory replay (Ours)	99.99 (AUC)	99.24

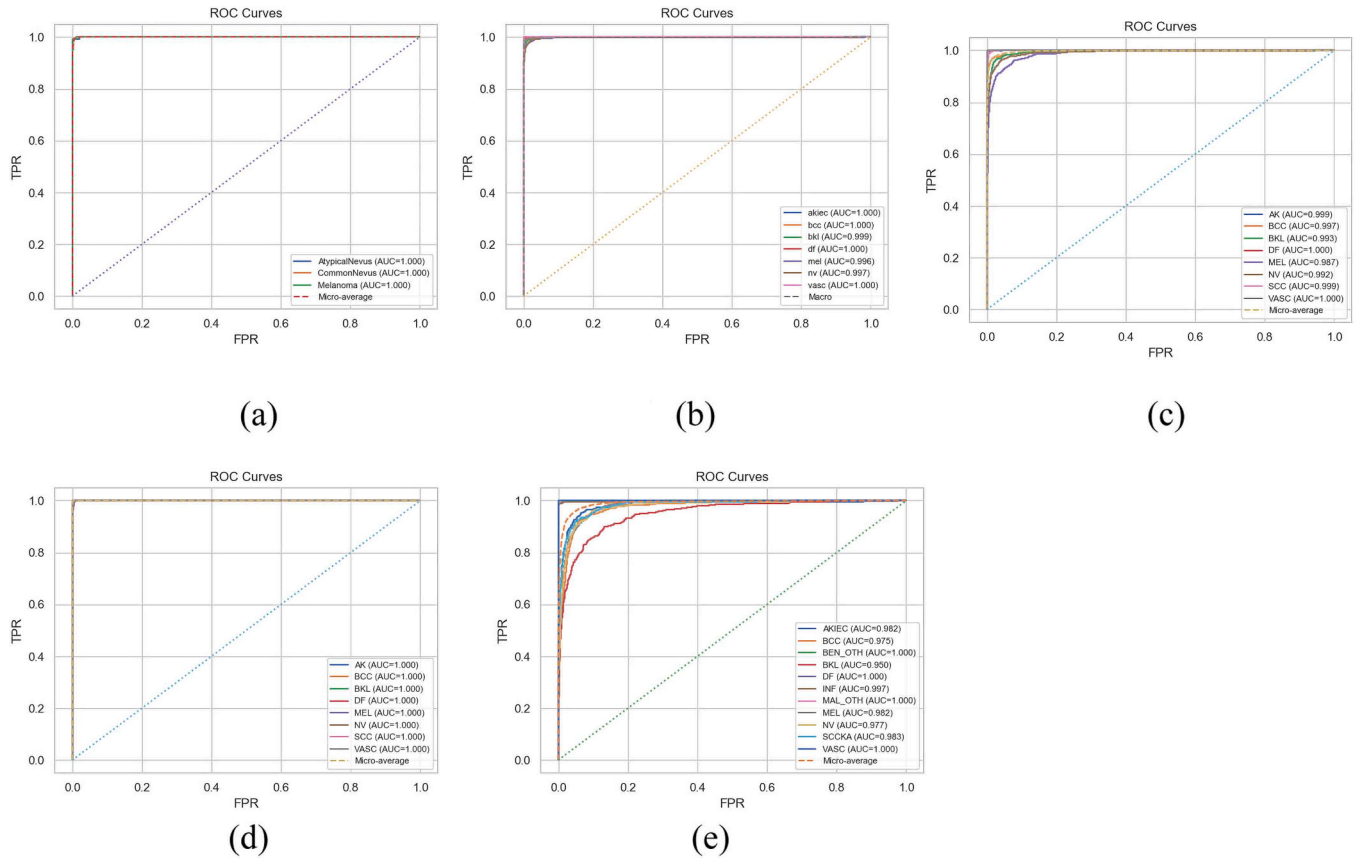


Fig. 4. ROC evaluating the diagnostic performance of proposed Agentic MobileViT framework across test sets of five benchmark skin cancer datasets demonstrating robust discrimination capability and clinical utility (a) PH2 (b) HAM10000 (c) ISIC2019 (d) ISIC2020 (e) MILK10k.

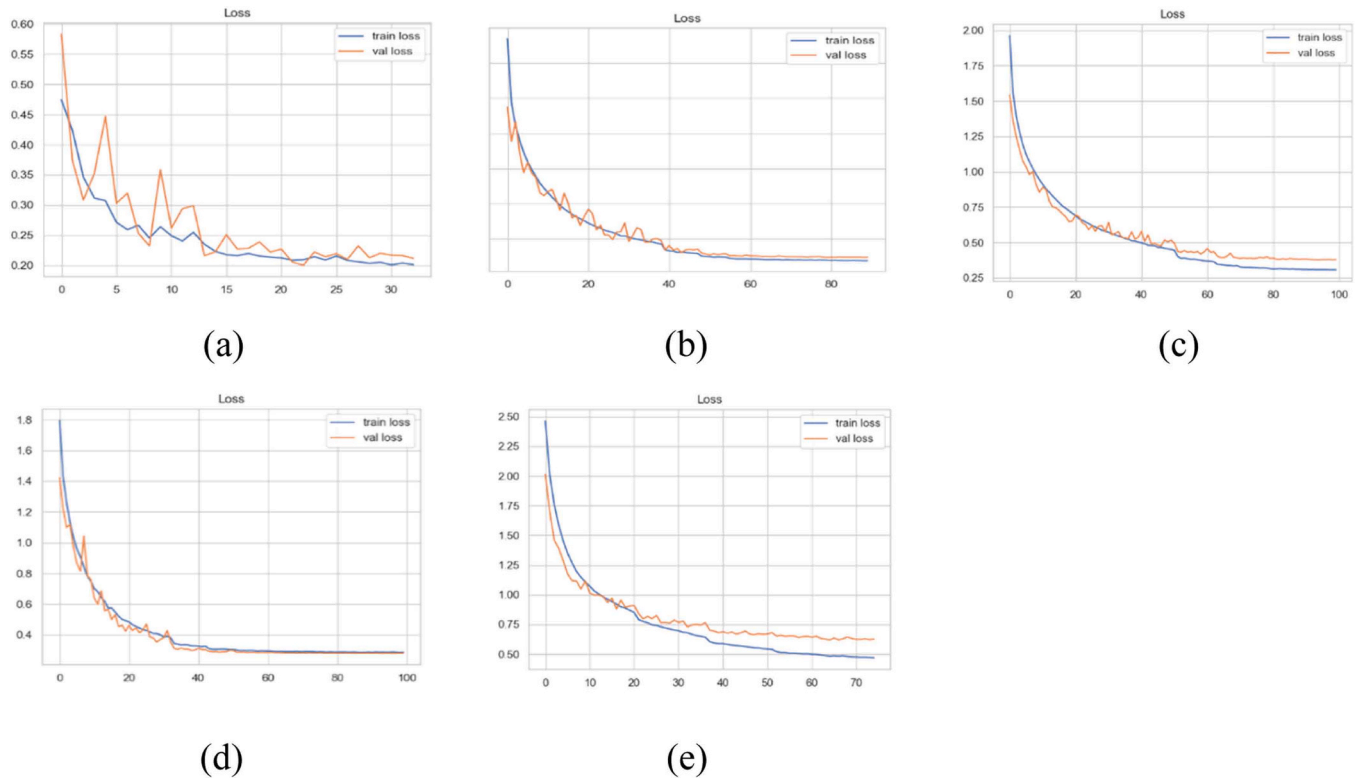


Fig. 5. Training loss curves representing loss per epochs and convergence behavior on five benchmark skin cancer datasets during incremental learning (a) PH2 (b) HAM10000 (c) ISIC2019 (d) ISIC2020 (e) MILK10k.

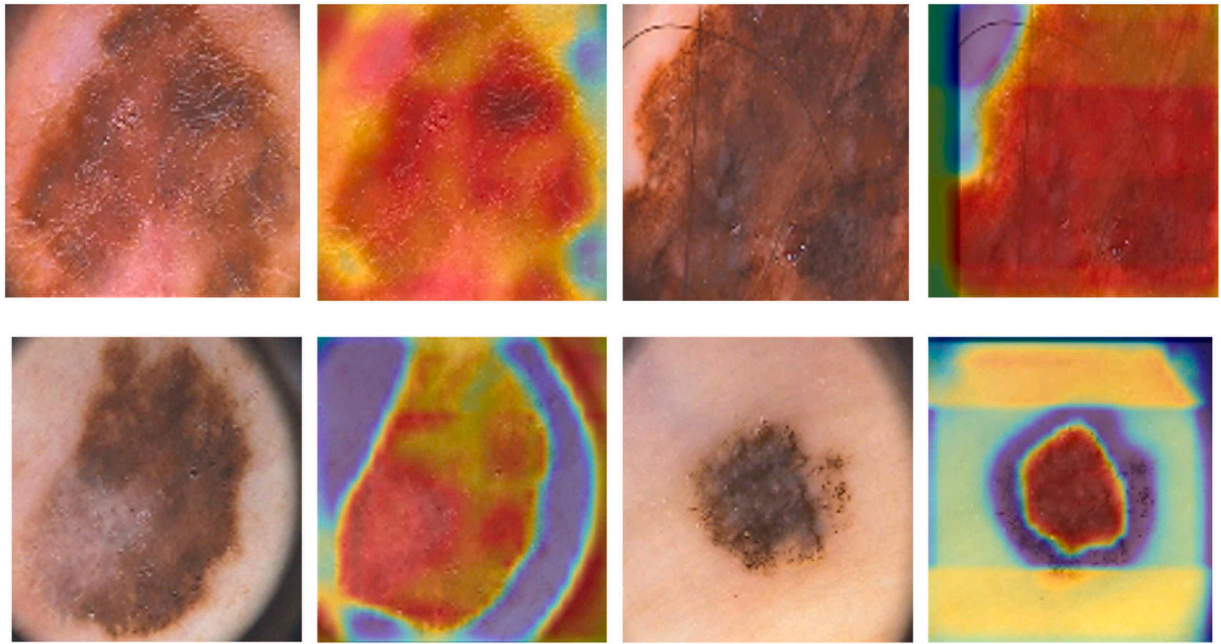


Fig. 6. Grad-CAM heatmap visualizations demonstrating the interpretability and clinical explainability of the proposed Agentic MobileViT framework's diagnostic decision-making process.

Table 5a

Comparison of agentic MobileViT with different backbones.

Dataset	Agentic MobileViT					Swin/EfficientNetViT (Test Acc)
	Train Acc	Val Acc	Test Acc	Inference (ms/img)	Params	
ISIC 2019	0.9910	0.9464	0.9448	2.55	1.68 M	67%-73%
ISIC 2020	0.9937	0.9920	0.9924	2.55	1.68 M	70%-75%
MILK10k	0.9616	0.8754	0.8706	2.55	1.68 M	65%-70%
PH2	0.9971	0.9836	0.9896	2.55	1.68 M	68%-74%
HAM10000	0.9984	0.9858	0.9811	2.55	1.68 M	65%-70%

RQ4. Our KD+Replay achieves lowest inference latency among all student variants at 2.55 ms/image, showing no increase in inference cost on addition of replay regularization addressing RQ3. Empirically, the training complexity follows $O\left(E \times \text{ceil}\left(\frac{N}{B}\right) \times C_{\text{fwd}+\text{bwd}}\right)$ where $E = 40, B = 24, E = 1760, \text{steps} = 74$. While the KD+Replay configuration shows 721.3 s, a higher training time compared to KD-only with 501.8 s, this overhead is only one-time training cost that has no effect on deployment inference latency consistent with the theoretical derivation.

While the backbone selection plays a crucial role in performance, another important factor influencing the success of proposed framework is using the Knowledge Distillation (KD) strategy. To further assess the

Table 5b

Comparison of agentic MobileViT computational efficiency across different model variants (PH2 dataset).

Model	Train Time (s)	Params	Inference (ms/image)	Val Acc	Test Acc
Teacher (EfficientNetB0)	83.0	4.38 M	3.23	98.18%	93.64%
Student only	348.7	1.68 M	3.56	80.00%	74.55%
KD only	501.8	1.68 M	6.04	79.09%	79.09%
KD+Replay (Proposed)	721.3	1.68 M	2.55	70.45%	74.09%
MobileNetV3Small	16.6	1.09 M	2.05	88.64%	80.91%
EfficientNetB0	28.6	4.38 M	3.27	96.82%	93.64%
ResNet50	38.9	24.1 M	3.01	95.45%	91.36%

independent contribution of each architectural element within MobileViT, an additional ablation study was performed on the PH2 dataset. All experiments were conducted on identical hardware, pre-processing pipeline, data splits and categorical cross-entropy loss to ensure fair comparison. We analyzed the performance of CNN only baseline, MobileViT without KD, and full MobileViT with KD.

The results reported in Table 6a address RQ2 and show that MobileViT with KD achieved a remarkably low loss of 0.010 with accuracy of 98.9% endorsing the robustness and strong generalization of the proposed framework. The performance drastically dropped when KD was removed yielding accuracy of only 61.2% and loss increasing to 0.780. A CNN-only baseline also achieved similar accuracy of 61.9% but a higher loss of 0.818. This suggests that CNN-only baseline captures local spatial features but struggles in generalization performance. The MobileViT baseline without KD introduces hybrid CNN-Transformer representations, achieving improved feature modeling, however training from scratch still results in moderate performance, whereas the full MobileViT baseline with KD shows considerable improvement in both loss and classification accuracy. This is the direct effect of knowledge transfer from pretrained teacher model. The integration of MobileViT's hybrid CNN-ViT backbone with KD not only improves discriminative feature and class separability but also improves convergence with reduced classification errors. In contrast, removing KD or relying only on CNN backbones lead to substantial degradation in model accuracy. The results suggest that the performance gain is only due to the architectural depth but is considerably influenced by KD during training process. The MobileViT student model addresses the elevated computational complexity of standard CNNs

through depthwise separable convolutions and local patch-based tokenization preventing the quadratic attention scaling that burdens full-resolution ViTs.

Fig. 7 presents the confusion matrices for the EfficientNetB0 teacher model on PH2 validation test data. It is significant to note that the melanoma class achieved a perfect recall with 0 misclassifications on both validation and test splits, ensuring that the teacher network provides soft targets for clinically most critical class. The identified 9 misclassifications of atypical nevus cases as common nevus on test data splits represent the primary source of classification error, which is consistent with the visual similarities between both classes.

To isolate the effect of proposed agentic optimizer switch (AOS), an ablation study was performed on PH2 dataset under identical experimental setup where we only changed the bounded agentic optimizer switch mechanism while keeping all other components of the proposed framework unchanged. We particularly analyzed and compared our full model with the AdamW→SGD agentic switch enabled against a variant where it was disabled.

Table 6b presents the results of ablation conducted to analyze the results with and without the AOS mechanism on PH2 dataset. The AOS-disabled configuration achieved an identical accuracy and recall of 0.9810, precision of 0.9814, and weighted F1-score of 0.9809 respectively, compared to 0.9833, 0.9840 and 0.9833 respectively on PH2 test set. This shows a consistent performance improvement of + 0.0026 in precision and + 0.0024 in accuracy, recall and F1-score upon enabling the bounded AOS mechanism. These performance gains, while incremental in absolute terms, ensure that the bounded AdamW→SGD transition meaningfully refines parameter convergence in later training epochs, consistent with its design as a convergence stabilizer rather than a primary performance driver. Enabling AOS increases the training time marginally from 913.26 to 974.12 s representing an increased overhead of approximately 6.7%. This additional cost is attributed to the stagnation monitoring and optimizer transition logic rather than any increase in processing cost per sample and only represents a one-time training overhead with no effect on the deployment. Remarkably, enabling AOS reduces the inference latency from 3.84 to 3.66ms/image showing an improvement of 4.7%. This suggests that a model with marginally more effective activation patterns at inference time is produced with more stable convergence achieved through AOS. The overall ablation results confirm that the AOS delivers consistent, incremental accuracy, and inference efficiency gains at a modest and acceptable training time cost, supporting its role as a lightweight convergence stabilizer rather than a primary performance driver.

Fig. 8 presents the confusion matrices for the AOS ablation performed on PH2 dataset test split. Model achieves 413 out of 420 correct classifications while AOS is enabled with perfect recall for common nevus and melanoma classes and only 7 atypical nevus misclassified as common nevus. In contrast, upon disabling AOS, the correct classifications drop to 412 with same 7 misclassifications for atypical nevus whereas 1 additional misclassification of common nevus as atypical nevus identified. Melanoma recall remains perfect in both cases which confirms that the AOS actually refines the discrimination boundary between visually similar benign categories without compromising the recall for the most critical melanoma class.

To summarize, this comprehensive comparative analysis confirms that MobileViT-based agentic architectures especially when enhanced with KD+replay and bounded optimizer switching are not only more reliable and robust but also more generalizable, and computationally

efficient for stage-wise progressive skin lesion classification across heterogeneous dermatological datasets. The results position these architectures as a promising backbone for future research in progressive multi-dataset medical image analysis where relying solely on traditional transformer or convolutional baselines may yield suboptimal adaptability under sequential training conditions.

5. Discussion

The findings from this work provide compelling evidence that the proposed Agentic MobileViT architecture offers a substantial development for stage-wise progressive skin lesion classification across heterogeneous multi-datasets, surpassing traditional convolutional and transformer-based baselines. The proposed architecture's core strength lies in its hybrid nature, which combines the local feature extraction through depth-wise separable convolutions with global contextual reasoning through transformer self-attention mechanisms.

The dual-path design in proposed model empowers it to capture both the fine-grained textures of lesions and broader contextual cues, contributing to its superior generalization across diverse skin disease datasets. Further, the ablation results validate that this hybrid architecture consistently outperforms Swin Transformer and EfficientNetViT, exceeding performance gain of over 20 %age points in several benchmarks, confirming that the model performance improvement is attributable to architectural integration rather than any isolated single component in the architecture.

A key innovation of the framework is the AMR mechanism, which effectively preserves representational knowledge across sequential training stages through class-importance-weighted prioritized sampling. The model effectively preserves the prior knowledge while adapting to new lesion categories by maintaining a balanced replay buffer of 1200 samples and jointly optimizing on both current and past data with ratio of 70:30. It is important to note that the AMR mechanism operates within a stage-wise progressive transfer learning paradigm rather than formal continual learning setting. Since all datasets are trained progressively with independent evaluation at the end of each stage, we do not compute the formal retention metrics such as forgetting scores and backward transfer in this study. The KD+ replay components are therefore best characterized as representation stabilization mechanism rather than catastrophic forgetting solutions in strict continual learning paradigm. This is an intentional design choice reflecting clinical reality of dermatological AI development where datasets from different institutions emerge incrementally and full joint retraining is computationally expensive.

Another important aspect of the proposed model is its dynamic optimizer switching using AOS which enhanced the framework's training robustness. This enables the system to adapt its optimization strategy upon detecting learning stagnation, thereby improving convergence stability across all five training stages without the need of dataset-specific hyperparameter tuning. Ablation results confirm its meaningful contribution to convergence speed while having a marginal effect on final classification accuracy.

Alignment of class label spaces across heterogeneous datasets is a critical methodological consideration in cross-dataset training. A global classifier head was constructed with output dimensionality compatible with union of all class labels across five datasets. Both teacher (EfficientNetB0) and student (MobileViT hybrid) models share the identical output dimensionality throughout training process. Consequently, the computed KD loss is defined over a consistent class space at each training iteration, without requiring any intermediate projection, class remapping or output masking. This unified classifier design resolves the class-alignment problem that usually arises in cross-dataset incremental learning and ensures that the distillation signal remains stable and informative across all training stages. However, while the fixed global classifier head prevents the output-space expansion and removes the logit-level class forgetting, it does not inherently protect the

Table 6a
Ablation study on PH2 dataset (Effect of Knowledge Distillation).

Model	Loss	Accuracy
CNN-only	0.818	0.619
MobileViT (No KD)	0.780	0.612
Full MobileViT+KD	0.010	0.989

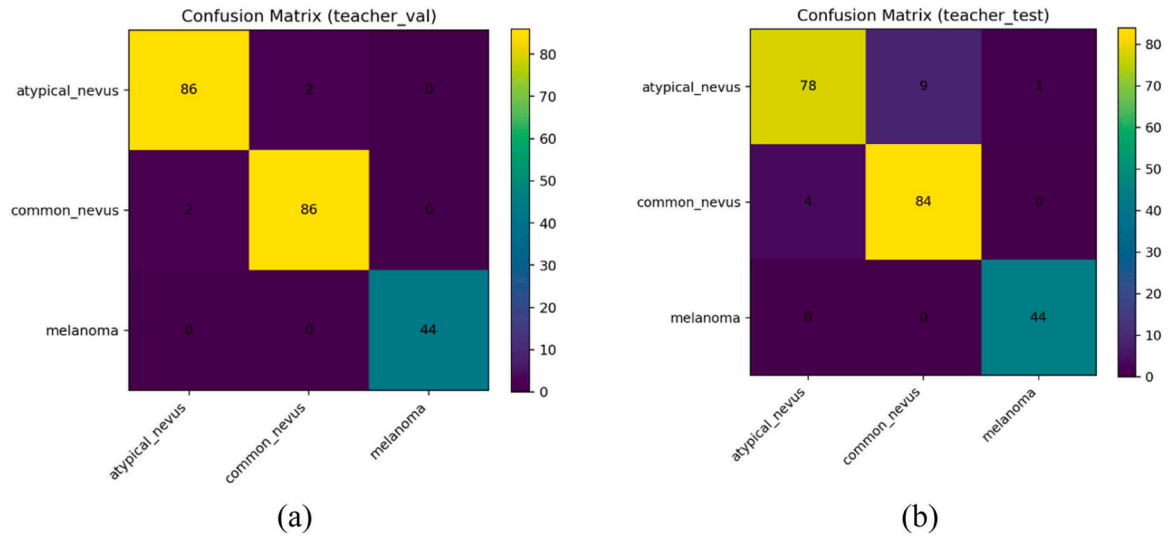


Fig. 7. Confusion matrices of EfficientNetB0 teacher model evaluated on PH2 dataset for (a) validation and (b) test splits where true and predicted class labels are represented in rows and columns, respectively across three lesion categories: atypical nevus, common nevus, and melanoma. Perfect melanoma recall on both splits confirm that the teacher model provides reliable soft targets for clinically most critical class. The primary source of misclassification reflects consistency with the visual overlap in atypical and common nevus class images.

Table 6b

Ablation study on PH2 dataset (Effect of Agentic Optimizer Switch).

AOS Setting	Val Accuracy	Test Accuracy & Recall	Test Precision	Test F1 score	Train time (sec)	Inference (ms/image)
OFF	0.97381	0.980952	0.981439	0.980946	913.2588	3.839341
ON	0.980952	0.983333	0.984	0.983326	974.1157	3.658815

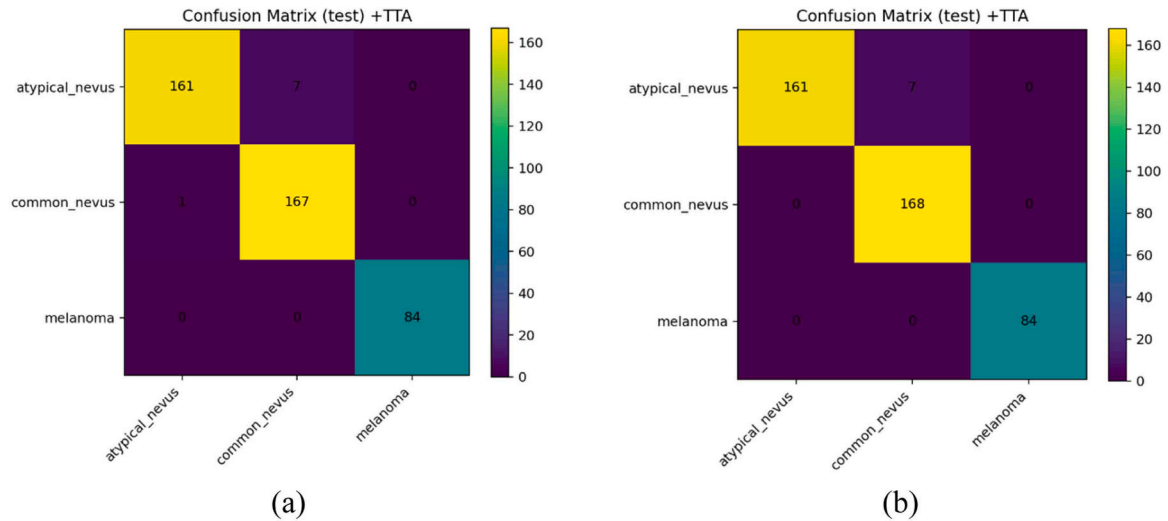


Fig. 8. Confusion matrices of proposed Agentic MobileViT evaluated on PH2 test set for (a) AOS disabled and (b) AOS enabled where true and predicted class labels are represented in rows and columns across three classes, atypical_nevus, common_nevus, and melanoma. Primary misclassification source remains the atypical_nevus and common_nevus due to visual overlap between the two benign categories.

intermediate feature representations from gradient interference during sequential fine-tuning. To address this limitation, KD and replay-based regularization serve as complimentary representation stabilizers. Specifically, KD constraints the student model's soft output distribution to continue being aligned with the teacher model's learned representations while the replay sampling re-exposes the model to prior data distributions at feature-level. The mechanisms therefore are not redundant under a fixed output space; rather they address a distinct and empirically confirmed failure model related to feature degradation ensuring the

knowledge retention across incremental learning stages.

Additionally, the model also demonstrated its clinical adaptability, consistently achieving accuracies above 94% with strong generalization across standard heterogeneous benchmark datasets such as newly published MILK10k. The model was tested across several datasets to highlight its potential to deal with real-world variability, and act as a robust and reliable diagnostic support tool in dermatology. Another key strength of the proposed work is the integration of Grad-CAM explainability as a formal phase of the pipeline. This ensures a transparent

decision making process by enabling the clinicians to verify whether the model is attending to diagnostically relevant regions of a lesion, which is a crucial step towards building confidence in AI-assisted medical systems.

5.1. Limitations

Regardless of the promising results, the model still has certain limitations that must be considered. The proposed framework was trained and evaluated on NVIDIA GeForce RTX 4060 GPU. While it is a widely accessible consumer-grade GPU, deployment in resource-constrained real-world clinical environments with limited infrastructure may require quantization, compression or KD to a lighter student model to meet real-time inference requirements. All five standard classification datasets used in our study for evaluation purpose are publically available standard datasets collected under specific imaging protocols and device configurations. This may not reflect generalizability to real-world clinical settings with diverse acquisition conditions, devices and population.

We measure the independent classification performance of proposed model on the held-out test split of each dataset after each training stage. Our evaluation protocol does not measure task-incremental retention which is a deliberate design choice as our framework treats each dataset as a progressive knowledge foundation rather than isolated task requiring post-sequence retention verification. We intend to incorporate formal retention metrics in future to provide a more comprehensive characterization of the proposed framework's cross-stage knowledge preservation.

It is crucial to discuss the limitation of this work in case of data splitting protocol applied to ISIC 2020 dataset, which is known to include multiple images from same patient. In our study, we applied dataset partitioning at image-level using a stratified ratio which does not explicitly enforce patient-disjoint separation across out splits. This may introduce a potential patient-level correlation risk between splits, where images from same patient may occur in both training and testing subsets. Our performance improvements reported in ablation and comparative analysis remain internally valid as all baselines and ablation configurations are evaluated under same splitting protocol. In future, a patient-disjoint splitting will be enforced to provide a stricter evaluation protocol.

Adding the replay memory while effective, introduces an additional storage overhead that can become heavy with large number of incremental steps. Future research will explore generative replay or adaptive memory compression to address this concern. Secondly, while the model exhibits strong generalization across standard skin datasets, its performance in real-world clinical environment with varying image acquisition protocols and devices requires rigorous validation. Although the work incorporated explainability providing visual insights using tools like Grad-CAM, they may not fully capture the underlying decision logic. Future work should also be directed toward exploring the multi-modal patient data integration, using more reliable interpretability methods to provide a complete understanding of the model's underlying decision logic. Another interesting future direction is to explore the metaheuristic hyperparameter optimization techniques as Modified Starfish Optimization Algorithm [54] and Puma Optimizers [55] to further enhance the framework's adaptive training capability.

6. Conclusion

This study demonstrates the powerful and adaptive nature of the proposed Agentic MobileViT, a stage-wise progressive transfer learning framework for multi-dataset skin lesion classification. Core innovation introduced by this work is its unique fusion of hybrid CNN-ViT architecture with dynamic knowledge distillation, replay-regularization and bounded agentic optimization techniques. The framework demonstrated exceptional classification capabilities across multiple heterogeneous

datasets (PH2, HAM10000, ISIC 2019, ISIC 2020, and MILK10k). On the clinically focused ISIC 2020 dataset, the model successfully achieved F1-score of 99.24% confirming its potential as a reliable diagnostic support tool. This work is also one of the first to test MILK10k dataset and provide a comprehensive performance report on this dataset founding a new key benchmark for future research. The model's robust generalization combined with explainable and interpretable predictions using Grad-CAM provides means in building trust in AI-assisted dermatological systems. The proposed Agentic MobileViT framework provides an interpretable, computationally efficient, and progressively adaptive foundation for real-world dermatological image analysis, setting a new benchmark for stage-wise progressive transfer learning in AI-driven medical diagnostics.

CRedit authorship contribution statement

Butt Wasir: Supervision, Formal analysis, Conceptualization. **Nazish Yousaf:** Writing – original draft, Methodology, Conceptualization. **amin javaria:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization.

Declaration of Competing Interest

All authors declare that there is no conflict of interest in this research

Data availability

No data was used for the research described in the article.

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